

Cyrus Tau

PhD Student in Biological and Biomedical Sciences · Harvard University
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EDUCATION

Harvard University August 2024 – present

PhD in Biological and Biomedical Sciences

Cambridge, MA · GPA: 3.83

- Thesis Advisor: Gordon Fishell, Ph.D
- Rotations: Zhixun Dou, Ph.D; Sandeep Robert Datta, MD, Ph.D; Gordon Fishell, Ph.D
- NSF Graduate Research Fellowship (2024–2026)
- Teaching Fellow, Genetics 201 (Fall 2025)

Relevant Coursework: Principles of Molecular Biology (BCMP 200), Principles of Genetics (GENETIC 201), Neuroimmune Interactions in Health and Disease (IMMUN 305QC), Audition: Neural Mechanisms, Perception and Cognition (SHBT 205), Genetics, Biochemistry and Physiology of Forebrain Inhibition (NEUROBIO 363L), Development, Stem Cells and Regeneration (CELLBIO 207), Emergent Computations Within Distributed Neural Circuits (MIT 9.530)

University of California, Berkeley August 2020 – May 2024

B.S. Bioengineering, Minor in Data Science

Berkeley, CA · GPA: 3.78

- Regents' and Chancellor's Scholar (highest scholarship awarded by the University)
- Dean's List, Fall 2022
- Department Chair's Award, UC Berkeley Bioengineering

Relevant Coursework: Molecular Biology (MCELLBI 110), Molecular Immunology (MCELLBI 150), Cell & Systems Biology (MCELLBI 130), Cellular & Molecular Neuroscience (MCELLBI 160), Circuits, Systems & Behavioral Neuroscience (MCELLBI 161), Introduction to Machine Learning (BIOENG 145), Biochemistry of Synthetic Biology (BIOENG 148), Practical Light Microscopy (BIOENG 168L), Biological Transport Phenomena (BIOENG 104), Instrumentation for Biology and Medicine (BIOENG 101), Neural Networks (COMPSCI 182), Structure and Interpretation of Computer Programs (COMPSCI 61A), Principles & Techniques of Data Science (DATA C100), Probability for Data Science (DATA C140), Biophysical Chemistry (CHEM C130), Honors Undergraduate Research (BIOENG H194)

AWARDS & HONORS

NSF Graduate Research Fellowship	2024–2026
Regents' and Chancellor's Scholar , University of California, Berkeley	2020–2024
Department Chair's Award , UC Berkeley Bioengineering	2024
Dean's List , University of California, Berkeley	Fall 2022
Oral Presentation Award , Annual Biomedical Research Conference for Minoritized Scientists	2023

RESEARCH EXPERIENCE

Fishell Lab — Harvard Medical School & Broad Institute

March 2025 – present

Graduate Student

Cambridge, MA

- Co-leading projects investigating autonomous and non-autonomous cues for inhibitory interneuron cell type determination

Datta Lab — Harvard Medical School

January 2025 – March 2025

Rotation Student

Boston, MA

- Contributed to a project uncovering the spatial code underlying olfactory receptor organization in the nose and brain

Dou Lab — Massachusetts General Hospital

October 2024 – December 2024

Rotation Student

Boston, MA

- Contributed to a project investigating histone dynamics in senescent cells

Hsu Lab — University of California, Berkeley

November 2020 – August 2024

Undergraduate Researcher

Berkeley, CA

- Co-led a project using CRISPR RNA targeting proteins to enable RNA writing; co-inventor on associated patent
- Won award at the Annual Biomedical Research Conference for Minoritized Scientists
- Performed experiments for a genome-wide CRISPR knockdown and activation screen to identify factors affecting SARS-CoV-2 infection
- Designed and tested novel amplification technologies for CRISPR-based diagnostics

Snyder Lab — Stanford University

October 2019 – January 2021

Research Assistant

Palo Alto, CA

- Assisted with executing and analyzing clinical trials studying the biology of depression and evaluating new treatment methods
- Prepared statistical analyses of trial results for publications and internal use

BioCurious

January 2018 – August 2020

Independent Researcher

Santa Clara, CA

- Designed and led a project to reduce immunogenicity of artificial red blood cells containing hemoglobin

PUBLICATIONS

Brann, D. *et al.* A spatial code governs olfactory receptor choice and aligns sensory maps in the nose and brain.

[CO-AUTHOR]

Cell **189**, 3358–3379.e30 (2026). doi:10.1016/j.cell.2026.03.051

Chandrasakeran, S.* , **Tau, C.*** *et al.* Rewriting endogenous human transcripts with dual CRISPR-guided 3' trans-splicing. [CO-FIRST AUTHOR]

Cell Systems (2025). doi:10.1016/j.cels.2025.101487

Biering, S.B. *et al.* Genome-wide bidirectional CRISPR screens identify mucins as host factors modulating SARS-CoV-2 infection. [CO-AUTHOR]

Nature Genetics **54**, 1078–1089 (2022). doi:10.1038/s41588-022-01131-x

Ganz, A.B. *et al.* Effects of an immersive psychosocial training program on depression and well-being: A randomized clinical trial. [CO-AUTHOR]

Journal of Psychiatric Research **150**, 292–299 (2022). doi:10.1016/j.jpsychires.2022.04.004

PATENTS

CRISPR RNA writing via trans-splicing. Co-inventor with P. Hsu, S. Chandrasakeran *et al.*

US Patent Application US20250283112A1. Related to the RESPLICE system for rewriting endogenous human transcripts using dual CRISPR-guided 3' trans-splicing.

TEACHING

Harvard University — Genetics 201

Fall 2025

Teaching Fellow

Cambridge, MA

– Teaching Fellow for graduate-level genetics course at Harvard University

Biomedical Engineering Society — UC Berkeley Chapter

2020 – 2022

Academic Co-chair

Berkeley, CA

– Organized a series of computational biology classes with over 30 students

ORAL PRESENTATIONS

Annual Biomedical Research Conference for Minoritized Scientists (ABRCMS)

November 2023

Phoenix, AZ

– Tau, C., Chandrasekaran, S., et al. “Programmable RNA Writing Using CRISPR Catalyzed Trans-Splicing”

– Won Oral Presentation Award

POSTER PRESENTATIONS

Inside IGI Health Conference

September 2023

Berkeley, CA

- Tau, C., Chandrasekaran, S., et al. “Programmable RNA Writing with CRISPR Cas13 Trans-Splicing”

UC-wide Bioengineering Conference

June 2023

Berkeley, CA

- Tau, C., Chandrasekaran, S., et al. “Programmable RNA Writing with CRISPR Cas13 Trans-Splicing”

COMPUTATIONAL SKILLS & PROJECTS

LANGUAGES

Python, R, Bash/Unix, Swift,
Objective-C, JavaScript

BIOINFORMATICS

RNA-seq Analysis, CRISPR Design,
STAR Alignment, Genomics

MACHINE LEARNING

Deep Learning, Reinforcement Learning,
Protein LLMs, GANs

RESPLICE

Computational tools for analyzing RESPLICE RNA-seq data. Scripts and pipelines for processing, aligning, and quantifying splicing patterns from the RESPLICE (RNA-guided trans-splicing with CRISPR) experimental system.

github.com/hsulab-arc/RESPLICE →

Detecting Riboswitches with Diffusion + SOFM

MIT 9.530 project combining generative and unsupervised learning to identify riboswitches from RNA sequences. Uses a diffusion model for structural predictions, graph embeddings, self-organizing maps to capture conformational diversity, and a classifier to distinguish genuine riboswitches from random sequences.

github.com/cyrustau21/murfalo_cyrustau_switch_project →

TCR Fine-tuning

Fine-tuning protein large language models (ProtT5) to develop comprehensive T-cell receptor embeddings. Includes LoRA fine-tuning, linear probing, UMAP visualization, and hierarchical clustering analysis of TCR sequences.

github.com/ArnNag/cs182-tcrfinetuning →

SERVICE & OUTREACH

Bioengineering High School Competition

2023 – 2024

Mentor

Berkeley, CA

- Mentored teams of high schoolers to develop proposals using machine learning for Parkinson's and allogeneic CAR T-cells
- Provided technical feedback and advice on pursuing higher education and research in STEM

Splash at Berkeley

November 2022

Class Co-teacher

Berkeley, CA

- Co-taught and co-developed curriculum for a class on genomic engineering, focusing on CRISPR-Cas9

Biomedical Engineering Society — UC Berkeley Chapter

September 2020 – May 2022

Academic Co-chair, Academic and Outreach Committee Member

Berkeley, CA

- Presented workshops on research in Bioengineering to disadvantaged middle and high school students
- Organized a series of computational biology classes with over 30 students
- Co-led a committee of 20 people; organized coffee chats, course registration workshops, and bi-weekly events

BioCurious

January 2018 – March 2020

Volunteer Instructor

Santa Clara, CA

- Created curriculum, taught, recruited, and trained volunteers to teach classes on essential skills such as PCR, protein purification, DNA transformation, gel electrophoresis, and machine learning

CONSULTING

MIT Alumni Angels of Northern California

January 2019 – January 2024

Biotech Technical/Business Consultant

Palo Alto, CA

- Planned and executed tutorial sessions to explain core technologies of presenting biotech startups to potential investors
- Consulted for companies that have raised over \$1M, including BioAmp Diagnostics, Cura Therapeutics, Melio Labs, and CRISP-HR Therapeutics

FOR FUN

DJ — Nucleocyde

- Producing and mixing electronic music under the name Nucleocyde

soundcloud.com/nucleocyde →

Photography

- Sharing photos on Adobe Portfolio

adobe.ly/4ff8zBJ →